

NCERT BIOLOGY CLASS 11 • UNIT 3

Academic Notes & Exam Preparation

CHAPTER 8: CELL — THE UNIT OF LIFE

Comprehensive • Detailed • Unabridged



Coverage: 100% Textbook Content



Focus: Cytology & Organelles



Approach: Reductionist Biology



Format: Landscape PDF Ready

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U3 Unit 3 & Reductionism

Bio G.N. Ramachandran

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Cellular Unity in Diversity

Biology is the study of living organisms. Detailed descriptions of form and appearance reveal diversity, but the **cell theory** emphasizes the *unity* underlying this diversity: the cellular organization of all life forms. Living phenomena require the integrity of cellular organization to be observed.

Reductionist Biology

Definition: The physico-chemical approach to study and understand living organisms by using cell-free systems to investigate processes in molecular terms. Concepts of physics and chemistry are applied to understand gross physiological processes (digestion, excretion, memory, defense, recognition) and diseased conditions.

Unit 3 Chapters

- **Chapter 8:** Cell: The Unit of Life (Structure & Organelles)
- **Chapter 9:** Biomolecules (Chemical analysis & organic compounds)
- **Chapter 10:** Cell Cycle and Cell Division (Mitosis & Meiosis)



G.N. Ramachandran (1922 – 2001)

Founder: 'Madras School' of Conformational Analysis

Early Life & Education

Born Oct 8, 1922 near Cochin (Kerala). Influenced by his father (math professor). Top-ranking B.Sc (Honors) Physics graduate from Univ. of Madras (1942). Received Ph.D. from Cambridge Univ. (1949).

Major Discoveries

- 1. Triple Helical Structure of Collagen:** Published in *Nature* (1954).
- 2. Ramachandran Plot:** Analysis of allowed conformations of proteins through peptide dihedral angles.

Linus Pauling Influence

Met Linus Pauling at Cambridge; deeply influenced by Pauling's models of alpha-helix and beta-sheet structures, which directed his focus to collagen structure. Passed away April 7, 2001 (aged 78).



CELL: THE UNIT OF LIFE

1. Foundations & Cell Theory

- **A.V. Leeuwenhoek:** Saw live cell
- **Robert Brown:** Discovered nucleus
- **Schleiden (1838):** Plant cells
- **Schwann (1839):** Animal cell/Plasma membrane
- **Virchow (1855):** *Omnis cellula-e cellula*

2. Overview & Cell Dimensions

- **Mycoplasma:** 0.3 μm (Smallest)
- **PPLO:** $\sim 0.1 \mu\text{m}$
- **Bacteria:** 3–5 μm (1–2 μm typical)
- **Human RBC:** 7.0 μm diameter
- **Ostrich Egg:** Largest isolated single cell

3. Prokaryotic Organization

- **Envelope:** Glycocalyx $\rightarrow$ Cell Wall $\rightarrow$ PM
- **Gram Stain:** Gram +ve vs Gram -ve
- **Mesosomes:** PM infoldings (Respiration/DNA)
- **Ribosomes:** 70S (50S + 30S) & Polysomes
- **Inclusions:** Phosphate, Cyanophycean, Glycogen

4. Eukaryotic Organelles

- **Endomembrane:** ER, Golgi, Lysosome, Vacuole
- **Energy:** Mitochondria & Plastids (70S, DNA)
- **Motility/Skeleton:** Cilia/Flagella (9+2), Centriole
- **Control:** Nucleus, Chromatin, Chromosomes

Fundamental Definition:

The cell is the **fundamental structural and functional unit of all living organisms**. Anything less than a complete structure of a cell does not ensure independent living.

Unicellular Organisms

Capable of:

1. Independent existence
2. Performing all essential functions of life

Multicellular Organisms

Composed of many cells working in a coordinated manner to form tissues, organs, and organ systems.

 Historical Milestones

- **Antonie Von Leeuwenhoek:** First saw and described a *live cell*.
- **Robert Brown:** Later discovered the *nucleus*.
- **Electron Microscope Invention:** Revealed all structural details of the cell.

Scientist & Year	Background	Key Findings & Contributions
Matthias Schleiden (1838)	German Botanist	Examined large number of plants; concluded all plants are composed of different kinds of cells forming plant tissues.
Theodore Schwann (1839)	German Zoologist	Studied animal cells; reported thin outer layer ('plasma membrane'). Stated cell wall is unique to plants. Proposed: animal & plant bodies are composed of cells and products of cells.
Rudolf Virchow (1855)	German Pathologist	First explained that cells divide and new cells arise from pre-existing cells (<i>Omnis cellula-e cellula</i>). Modified Schleiden & Schwann hypothesis to finalize Cell Theory.

★ **Modern Cell Theory Postulates (Final Shape):**

1. All living organisms are composed of cells and products of cells.
2. All cells arise from pre-existing cells (*Omnis cellula-e cellula*).

 Onion Peel Cell (Typical Plant Cell)

- Outer boundary: Distinct **Cell Wall**.
- Inner boundary: **Cell Membrane** located just within the cell wall.
- Possesses a large central vacuole, plastids, and dense membrane-bound nucleus.

 Human Cheek Cell (Typical Animal Cell)

- Delimiting structure: Outer **Cell Membrane** (No Cell Wall).
- Contains dense membrane-bound nucleus and centrioles.
- Main arena of cellular activities: **Cytoplasm** (semi-fluid matrix).

 COMPLETE NCERT CELL SIZE & SHAPE REFERENCE MATRIX (FIGURE 8.1 DATA)

Cell Type / Organism	Dimension / Size	Shape Characteristics	Key Biological Feature
Mycoplasma	0.3 μm length	Pleomorphic (Variable shape)	Smallest known cell; lacks cell wall entirely.
PPLO (Pleuro Pneumonia Like Org.)	About 0.1 μm	Irregular / Pleomorphic	Extremely minute pathogenic prokaryote.
Typical Bacteria	1–2 μm (Gen. 3–5 μm)	Bacillus, Coccus, Vibrio, Spirillum	Prokaryotic organization, rapid multiplication.
Ostrich Egg	Largest isolated single cell	Spherical / Oval	Macroscopic single cell.
Human Red Blood Cells (RBCs)	7.0 μm in diameter	Round and Biconcave	Lacks nucleus at maturity; optimizes oxygen delivery.
White Blood Cells (WBCs)	Variable (10–20 μm)	Amoeboid	Can squeeze through capillary walls (diapedesis).
Columnar Epithelial Cells	Variable	Long and Narrow	Found in intestinal lining; absorption/secretion.
Nerve Cell (Neuron)	Some of the longest cells	Branched and Long	Conduction of nerve impulses across distance.
Tracheid	Elongated	Elongated with tapered ends	Xylem element; transport of water and support.
Mesophyll Cells	Variable	Round and Oval	Photosynthetic leaf tissue; packed with chloroplasts.

Cell Type / Organism	Dimension / Size	Shape Characteristics	Key Biological Feature
Typical Eukaryotic Cell	10–20 μm	Varied (Spherical, Polygonal, Cuboid)	Extensive cytoplasmic compartmentalization.
Viruses	0.02–0.2 μm	Helical, Polyhedral, Complex	Non-cellular obligate parasites.

Representatives & Features

Represented by **bacteria**, **blue–green algae (cyanobacteria)**, **mycoplasma**, and **PPLO**. Generally smaller and multiply much more rapidly than eukaryotic cells.

Cell Wall: Present in all prokaryotes except *Mycoplasma*.

4 Basic Shapes of Bacteria

Bacillus Rod-like shape

Coccus Spherical shape

Vibrio Comma-shaped

Spirillum Spiral shape

Genetic Material & Plasmids

Naked DNA: Not enveloped by a nuclear membrane. Consists of a single circular chromosome (Genomic DNA).

Plasmids: Small circular DNA outside genomic DNA. Confers unique phenotypic characters like **antibiotic resistance**. Used in monitoring bacterial transformation.

8.4.1 CELL ENVELOPE AND ITS MODIFICATIONS

LAYER 1 (OUTERMOST)

Glycocalyx

Differs in composition and thickness among bacteria:

- **Slime Layer:** Loose sheath in some bacteria.
- **Capsule:** Thick and tough layer in others.

LAYER 2 (MIDDLE)

Cell Wall

- Determines the shape of the cell.
- Provides strong structural support.
- Prevents bacterium from **bursting or collapsing**.

LAYER 3 (INNERMOST)

Plasma Membrane

- Selectively permeable in nature.
- Interacts with the outside world.
- Structurally similar to eukaryotic plasma membrane.

GRAM STAINING CLASSIFICATION (CHRISTIAN GRAM PROCEDURE)

Differential Staining

Feature	Gram Positive Bacteria	Gram Negative Bacteria
Gram Stain Retention	Retains crystal violet / iodine stain (Appears purple/blue).	Does not retain crystal violet stain; takes counterstain safranin (Appears pink/red).
Cell Envelope Structure	Thick peptidoglycan cell wall; single layer, smooth.	Complex multi-layered cell envelope with outer lipopolysaccharide membrane.

MESOSOMES & CHROMATOPHORES

Mesosome: Infoldings of cell membrane formed as **vesicles, tubules, and lamellae**.

- Cell wall formation
- DNA replication and distribution to daughter cells
- Respiration and secretion processes
- Increases surface area of plasma membrane & enzymatic content

Chromatophores: Membranous extensions in cytoplasm of cyanobacteria containing photosynthetic pigments.

FLAGELLA, PILI & FIMBRIAE

Bacterial Flagellum (Motility): Thin filamentous extension composed of 3 parts:

1. **Filament:** Longest portion extending outside.
2. **Hook:** Curved tubular connector.

3. **Basal Body:** Anchors flagellum into cell wall/membrane.

Pili: Elongated tubular structures made of special protein (pilin). (*No role in motility*)

Fimbriae: Small bristle-like fibers sprouting out; help attach bacteria to rocks in streams & host tissues.

8.4.2 RIBOSOMES AND INCLUSION BODIES

Prokaryotic Ribosomes (70S)

- Associated with plasma membrane of the cell.
- Dimensions: **15 nm by 20 nm** in size.
- Made of 2 subunits: **50S (large)** and **30S (small)** → forms **70S**.
- Primary site of **protein synthesis**.

Polyribosomes / Polysome: Several ribosomes attached to a single mRNA chain. Translate mRNA into proteins.

Inclusion Bodies (Reserve Materials)

- Non-membrane bound structures lying free in the cytoplasm.
- **Phosphate granules**
- **Cyanophycean granules**
- **Glycogen granules**

Gas Vacuoles: Found in blue-green, purple, and green photosynthetic bacteria (provides buoyancy).

Eukaryotes include all **protists, plants, animals, and fungi**. Eukaryotic cells exhibit extensive compartmentalization of cytoplasm through membrane-bound organelles, an organized nucleus with nuclear envelope, complex locomotory and cytoskeletal structures, and genetic material organized into chromosomes.

 **PLANT CELL VS ANIMAL CELL COMPLETE COMPARISON TABLE (FIGURE 8.3 DATA)**

Structural Feature	Plant Cell (Fig 8.3a)	Animal Cell (Fig 8.3b)
Cell Wall & Middle Lamella	Present (Rigid outer boundary made of cellulose, pectins, etc.)	Absent (Delimited only by plasma membrane).
Plastids (Chloroplasts, etc.)	Present (Essential for photosynthesis and storage).	Absent.
Central Vacuole	Large single central vacuole occupying up to 90% cell volume.	Vacuoles are small, temporary, or absent.
Centrosome / Centrioles	Absent in almost all plant cells.	Present (Helps in spindle formation during cell division).
Plasmodesmata	Present (Connects cytoplasm of adjacent plant cells).	Absent (Intercellular junctions like tight/gap junctions present).
Microvilli	Absent.	Present in specialized absorptive cells (e.g., gut epithelial cells).
Shared Organelles	Both possess Nucleus, Nucleolus, Nuclear Envelope, Rough ER, Smooth ER, Golgi Apparatus, Mitochondria, Peroxisomes, Lysosomes, Microtubules, Cytoplasm, and 80S Ribosomes.	

Chemical Composition

Studied extensively in **human red blood cells (erythrocytes)**:

52%
Protein

40%
Lipids

Also contains carbohydrates and **cholesterol**.

Phospholipid Bilayer

- **Polar Head:** Hydrophilic; directed towards the outer sides.
- **Hydrophobic Tails:** Saturated hydrocarbons; directed towards inner part.
- Protects nonpolar tails from aqueous environment.

Membrane Proteins (Ease of Extraction)

- **Peripheral Proteins:** Lie on the surface of the membrane.
- **Integral Proteins:** Partially or totally buried in the membrane bilayer.



Singer and Nicolson (1972) — Fluid Mosaic Model

Widely Accepted Model

Proposed that the **quasi-fluid nature of lipid** enables lateral movement of proteins within the overall bilayer. This ability to move within the membrane is measured as its **fluidity**.

Cell Growth

Intercellular Junctions

Secretion

Endocytosis

Cell Division

Transport Type	Energy Requirement	Direction of Movement	Mechanism & Examples
Passive Transport (Simple Diffusion)	No Energy (ATP-independent)	Along concentration gradient (Higher → Lower)	Neutral solutes move across without carrier proteins.
Osmosis	No Energy (ATP-independent)	Along water concentration gradient	Movement of water by diffusion across selectively permeable membrane.
Facilitated Transport	No Energy (ATP-independent)	Along concentration gradient	Polar molecules cannot pass nonpolar lipid bilayer; require carrier protein .
Active Transport	Energy Required (ATP utilized)	Against concentration gradient (Lower → Higher)	Example: Na⁺/K⁺ Pump .

Definition & Functions

Non-living rigid outer covering for plasma membrane of fungi and plants.

- Gives shape to the cell
- Protects from mechanical damage and infection
- Helps in cell-to-cell interaction
- Barrier to undesirable macromolecules

Chemical Composition

- **Algae Cell Wall:** Cellulose, galactans, mannans, and minerals like calcium carbonate (CaCO_3).
- **Plant Cell Wall:** Cellulose, hemicellulose, pectins, and proteins.

Wall Layers & Plasmodesmata

- **Primary Wall:** Capable of growth in young plant cell; diminishes as cell matures.
- **Secondary Wall:** Formed on inner side (towards membrane) as cell matures.
- **Middle Lamella:** Made mainly of **calcium pectate**; glues neighbouring cells together.
- **Plasmodesmata:** Cytoplasmic connections traversing cell wall & middle lamellae.

 Endomembrane Definition & Inclusion Rule:

Membranous organelles whose functions are **coordinated** form the Endomembrane System.

1. Endoplasmic Reticulum (ER)

2. Golgi Apparatus

3. Lysosomes

4. Vacuoles

 **EXCLUDED ORGANELLES:** Mitochondria, Chloroplasts, and Peroxisomes are NOT part of the endomembrane system because their functions are not coordinated with the components above!

 8.5.3.1 Endoplasmic Reticulum (ER)

Fig 8.5

Network of tiny tubular structures scattered in cytoplasm. Divides intracellular space into two compartments:

1. **Luminal Compartment:** Inside ER.
2. **Extra-luminal Compartment:** Cytoplasm.

Rough ER (RER)

Bears ribosomes on surface. Active in **protein synthesis and secretion**. Continuous with outer nuclear membrane.

Smooth ER (SER)

Lacks ribosomes. Major site for **lipid synthesis**. In animal cells, synthesizes **steroidal hormones**.

 8.5.3.2 Golgi Apparatus

Camillo Golgi (1898)

Densely stained reticular structure near nucleus. Consists of flat, disc-shaped sacs/cisternae of **0.5 μm – 1.0 μm diameter** stacked parallel.

Concentric Faces:

- **Cis / Forming Face:** Convex, faces ER (receives vesicles).
- **Trans / Maturing Face:** Concave (releases modified vesicles).

Functions: Packaging materials; major site of formation of **glycoproteins and glycolipids**.

8.5.3.3 Lysosomes

Membrane-bound vesicular structures formed by packaging in Golgi apparatus.

Hydrolytic Enzymes (Hydrolases):

Extremely rich in **lipases**, **proteases**, **carbohydrases** optimally active at **acidic pH**. Capable of digesting carbohydrates, proteins, lipids, and nucleic acids.

8.5.3.4 Vacuoles

Membrane-bound space in cytoplasm containing water, sap, excretory products. Bound by single membrane called **tonoplast**. Occupies up to **90% cell volume** in plants.

- **Tonoplast Transport:** Facilitates ion transport *against concentration gradient* into vacuole (concentration much higher in vacuole than cytoplasm).
- **Amoeba:** *Contractile vacuole* for osmoregulation & excretion.
- **Protists:** *Food vacuoles* formed by engulfing food.

Morphology & Dimensions

- **Shape:** Sausage-shaped or cylindrical.
- **Diameter:** 0.2–1.0 μm (Average: **0.5 μm**).
- **Length:** 1.0–4.1 μm .
- **Visibility:** Not easily visible unless specifically stained. Number varies per cell based on physiological activity.

Double Membrane & Cristae

- **Outer Membrane:** Forms continuous limiting boundary.
- **Inner Membrane:** Forms infoldings called **cristae** towards matrix (increases surface area).
- **Inter-membrane Space:** Outer compartment between membranes.

Matrix & Function

- Site of **aerobic respiration**; produces ATP.
- **Matrix contains:** Single circular DNA, few RNA molecules, **70S ribosomes**, protein synthesis components.
- Division: Divide by **fission**.

 CLASSIFICATION OF PLASTIDS (BASED ON PIGMENTS & STORED NUTRIENTS)**1. Chloroplasts**

Contain **chlorophyll** and **carotenoid pigments**. Responsible for trapping light energy essential for photosynthesis.

2. Chromoplasts

Contain fat-soluble carotenoid pigments like **carotene**, **xanthophylls**. Imparts yellow, orange, or red color to plant parts.

3. Leucoplasts (Colourless)

- **Amyloplasts**: Store carbohydrates (starch), e.g., potato.
- **Elaioplasts**: Store oils and fats.
- **Aleuroplasts**: Store proteins.

 Chloroplast Architectural Details (Mesophyll Cell Plastid)

Fig 8.8 Data

Number & Shape

Shapes: Lens, oval, spherical, discoid, ribbon-like.

Number: **1 per cell** in *Chlamydomonas*; **20 – 40 per cell** in mesophyll.

Dimensions

Length: **5–10 μm**

Width: **2–4 μm**

Double Membrane

Inner membrane is relatively *less permeable*. Space enclosed by inner membrane is called **Stroma**.

Thylakoids & Grana

Thylakoids: Flattened sacs stacked like coins into **grana**. Connected by **stroma lamellae**.

Stroma Contents: Contains enzymes for carbohydrate/protein synthesis, small double-stranded circular DNA, **70S ribosomes** (smaller than 80S cytoplasmic ribosomes), and chlorophyll pigments enclosed inside thylakoid lumens.

8.5.6 RIBOSOMES (80S vs 70S Comparison)

George Palade (1953) • Fig 8.9

First observed under electron microscope as dense particles by **George Palade (1953)**. Composed of **ribonucleic acid (RNA) and proteins**; not surrounded by any membrane.

Eukaryotic Ribosome (80S)

Cytoplasmic

- Large Subunit: 60S
- Small Subunit: 40S

Note: Sedimentation coefficients do not add up linearly ($60S + 40S = 80S$).

Prokaryotic / Organellar Ribosome (70S)

Bacteria, Mito, Chloroplast

- Large Subunit: 50S
- Small Subunit: 30S

Found in prokaryotes and inside mitochondria and chloroplasts ($50S + 30S = 70S$).

'S' (Svedberg's Unit): Stands for the sedimentation coefficient; indirectly a measure of density and size.

An elaborate network of filamentous proteinaceous structures present in the cytoplasm.

3 Main Components:

1. **Microtubules** (Tubulin protein)
2. **Microfilaments** (Actin protein)
3. **Intermediate Filaments**

Primary Functions:

Mechanical Support

Motility

Maintenance of Cell Shape

Hair-like outgrowths of cell membrane. **Cilia** are small (work like oars); **Flagella** are longer (responsible for cell movement).

Internal Structure of Axoneme (9+2 Array):

- **9 Peripheral Microtubule Doublets** arranged radially.
- **1 Pair of Centrally Located Microtubules** (connected by bridges & enclosed by central sheath).
- **9 Radial Spokes:** Connect central sheath to peripheral doublets.
- **Interdoublet Linkers:** Connect adjacent peripheral doublets.
- Emerge from centriole-like structure called **Basal Body**.

Organization

Organelle containing **two cylindrical structures called centrioles** lying perpendicular to each other, surrounded by amorphous pericentriolar materials.

Cartwheel Architecture

- Made of **9 evenly spaced peripheral triplets** of tubulin protein.
- Central proteinaceous **Hub** connected to peripheral triplets by proteinaceous **radial spokes**.

Functions

- Forms the **basal body** of cilia and flagella.
- Forms **spindle apparatus** during cell division in animal cells.

Discovery & Envelope

- First described by **Robert Brown (1831)**.
- Stained with basic dyes named **Chromatin** by **Flemming**.
- **Nuclear Envelope**: 2 parallel membranes with **10–50 nm perinuclear space**. Interrupted by nuclear pores.

Nucleolus & DNA Packaging

- **Nucleolus**: Non-membrane bound spherical body in nucleoplasm; site for active **rRNA synthesis**.
- **Human Cell DNA**: Single human cell has approx. **2 metre long DNA** distributed among **46 (23 pairs) chromosomes**.
- Contains DNA, basic proteins (**histones**), non-histones & RNA.

Special Enucleate Cells

- **Mature Mammalian RBCs**: Lack nucleus.
- **Sieve Tube Cells**: Vascular plant cells lacking nucleus at maturity.

 CLASSIFICATION OF CHROMOSOMES BASED ON CENTROMERE POSITION (FIGURE 8.13)

Chromosome Type	Centromere Position	Arm Length Ratio	Visual Appearance during Anaphase
Metacentric	Middle centromere	Two equal arms	'V'-shaped
Sub-metacentric	Centromere slightly away from middle	One shorter arm (p arm) + One longer arm (q arm)	'L'-shaped
Acrocentric	Centromere situated close to its end	One extremely short arm + One very long arm	'J'-shaped
Telocentric	Terminal centromere	Only one arm visible (Centromere at tip)	'I'-shaped

Secondary Constriction & Satellite: Sometimes a few chromosomes have non-staining secondary constrictions at a constant location, giving the appearance of a small fragment called a **satellite**.

**SAT-
Chromosome**

Many membrane-bound minute vesicles called **microbodies** (e.g., peroxisomes containing catalase and oxidase enzymes) that contain various enzymes are present in **both plant and animal cells**.

CHROMOSOME SHAPES**Centromere Position Types**

Mnemonic: **M-S-A-T** ("My Sister Acts Today")

- **Metacentric** (Middle / Equal arms)
- **Sub-metacentric** (Slightly away)
- **Acrocentric** (Almost at end)
- **Telocentric** (Terminal end)

LEUCOPLAST TYPES**Stored Nutrient Types**

Mnemonic: **A-E-A** ("Amy Eats Ale")

- **Amyloplasts**: Starch / Carbohydrates
- **Elaioplasts**: Essential Oils & Fats
- **Aleuroplasts**: Amino acids / Proteins

ENDOMEMBRANE SYSTEM**Included Components**

Mnemonic: **GERL** ("Golgi-ER-Ribosome-Free Lysosome-Vacuole")

- **G**: Golgi Apparatus
- **E**: Endoplasmic Reticulum
- **R/L**: Lysosome
- **V**: Vacuole

FINAL REVISION & NUMERICAL METRICS MATRIX

Quick Revision Sheet

0.3 μm

Mycoplasma Length

0.1 μm

PPLO Size

7.0 μm

Human RBC Diameter

15 x 20 nm

Ribosome Size

0.5 – 1.0 μm

Golgi Cisternae Diameter

1.0 – 4.1 μm

Mitochondria Length

5 – 10 μm

Chloroplast Length

2 – 4 μm

Chloroplast Width

10 – 50 nm

Perinuclear Space

2 Metres

Human Cell DNA Length

52% / 40%

RBC Protein / Lipid Ratio

9 + 2

Cilia/Flagella Array